

Practical Numerics for hybrid systems

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Abstract

Simulating a hybrid trajectory generally requires writing problem-specific solvers. This is due to the fact that many different systems can be classified as hybrid. For example, a Filippov system has continuous trajectories but detection for sliding, crossing, and escaping need to be considered. On the other hand, a mechanical impact system admits discontinuous trajectories and can experience arbitrarily frequent impacts/Zeno trajectories. Both of these systems require qualitatively different techniques to obtain numerical solutions.

We present a software package that encompasses a wide variety of hybrid systems and automatically applies the correct stepper logic. In particular, sliding mode is implemented for Filippov systems, singular resets are detected for discontinuous hybrid systems, and nonholonomic constraints and post-Zeno extension can be applied to mechanical systems.

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